

# Project 2 – Pneumonia Detection Using CNN

## 1. Introduction

Pneumonia is a lung infection that can be dangerous if not detected early. Chest X-ray images are commonly used for diagnosis, but manual examination can be slow and depends on medical expertise. Convolutional Neural Networks (CNNs) are well suited for this task because they can automatically learn important visual features from images.

The goal of this project is to build a CNN model that classifies chest X-ray images into **Normal** or **Pneumonia**.

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## 2. Dataset

The model was trained using a public **Chest X-ray Pneumonia dataset**. The dataset contains chest X-ray images labeled as:

- **NORMAL**
- **PNEUMONIA**

The data is divided into training, validation, and testing sets. The dataset is imbalanced, with more pneumonia images than normal images, which makes recall an important evaluation metric.

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### 3. Data Preprocessing

Before training, the following preprocessing steps were applied:

- Images were resized to a fixed input size
- Pixel values were normalized by dividing by 255 to make the range of (0 to 1)
- Images were loaded in batches
- Labels were encoded for binary classification

These steps help the CNN train more efficiently and generalize better.

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### 4. CNN Model Architecture

A custom CNN model was built using Keras with the following structure:

- Three convolutional layers with 32, 64, and 128 filters
  - Max pooling layers after each convolution
  - A fully connected layer with 128 neurons
  - Dropout layer to reduce overfitting
  - Final output layer for binary classification
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## 5. Training Setup

- **Optimizer:** Adam
- **Loss function:** Binary Cross-Entropy
- **Epochs:** Up to 20
- **Early stopping:** Enabled
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- **Learning rate reduction:** Applied during training

The best model was saved automatically based on validation loss.

**Callbacks were used in the training**

### What are callbacks (in Keras or TensorFlow)?

**Callbacks are tools that watch the training process and automatically take actions while the model is training.**

## Callbacks used in the training loop:

### 1) EarlyStopping

```
tf.keras.callbacks.EarlyStopping(  
    monitor="val_loss",  
    patience=6,  
    restore_best_weights=True,  
    verbose=1  
)
```

Monitors validation loss

If `val_loss` does not improve for 6 epochs, training stops automatically

Restores the best model weights

helps in:

Stops training before overfitting

Prevents wasting time on useless epochs

## 2) ReduceLROnPlateau

```
tf.keras.callbacks.ReduceLROnPlateau(  
    monitor="val_loss",  
    factor=0.5,  
    patience=3,  
    min_lr=1e-6,  
    verbose=1  
)
```

Watches `val_loss`

If `val_loss` stops improving for 3 epochs

The learning rate is reduced by half

Help in :

Large learning rates are good at the start

Smaller learning rates help fine-tune the model later

Helps the model escape plateaus

### 3) ModelCheckpoint

```
tf.keras.callbacks.ModelCheckpoint(  
    "pneumonia_best_model.keras",  
    monitor="val_loss",  
    save_best_only=True,  
    verbose=1  
)
```

Saves the model **only when validation loss improves**

Keeps the **best model**, not the last one

#### Why this matters:

The last epoch is **not always the best**

Always get the **best-performing model**

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## 6. Results and Evaluation

### Training Performance

The model showed steady improvement during training indicating good learning performance.

### Test Results

The confusion matrix on the test set showed:

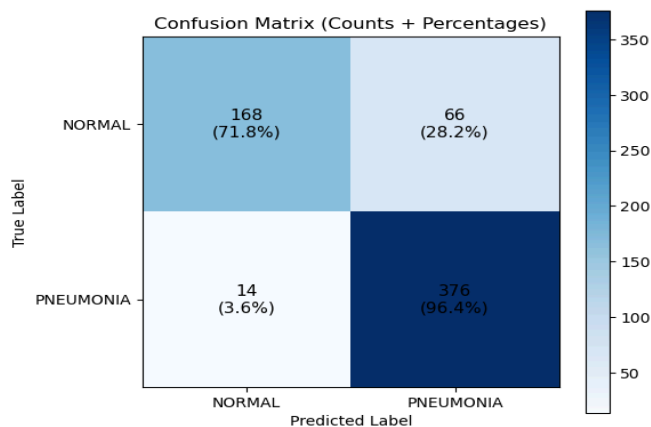
- Correct detection of most pneumonia cases
- Some normal cases were incorrectly classified as pneumonia

**Test accuracy: ~87%**

**Key metrics:**

- **Pneumonia recall: ~96%**
- **Normal recall: ~72%**

This means the model is very effective at detecting pneumonia, which is important in medical screening tasks.



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## 7. Conclusion

A CNN model was successfully developed to detect pneumonia from chest X-ray images. The model achieved high validation accuracy and strong pneumonia detection performance. Although some normal images were misclassified, the high pneumonia recall makes the model suitable as a supportive diagnostic tool.

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## 8. Future Work

Possible improvements include:

- Using pre-trained CNN models (transfer learning like ResNet, VGG on ImageNet)
- Reducing model size to improve generalization
- Applying class balancing techniques